Obstetric & Gynecologic Critical Care – Chapter Summary

Physiology • Hypertension • Hemorrhage • Emergencies

ICU / EDIC Board Review • MCQ-Oriented

1. Physiology of Pregnancy

- **CO ↑ 40–50%** (HR ↑, SV ↑); SVR ↓; BP lowest in 2nd trimester
- **Blood volume ↑ 40–50%**, plasma > RBC → dilutional anemia
- **Minute ventilation ↑ 40–50%** (progesterone) → **$PaCO_2$ 28–32**, compensated resp alkalosis (HCO_3 18–21)
- **FRC ↓ 20%**, O_2 consumption ↑ → rapid desaturation
- **Hypercoagulable:** ↑ fibrinogen (> 4 g/L), ↑ factors VII, VIII, X
- **GFR ↑ 50%** → creatinine normally 0.4–0.8 mg/dL
- **Aortocaval compression** > 20 weeks → left lateral tilt / manual uterine displacement

$PaCO_2$ 40 in pregnancy = impending respiratory failure

2. Hypertensive Disorders

- **Preeclampsia:** BP ≥ 140/90 after 20 weeks + proteinuria (≥ 300 mg/24 h, PCR ≥ 0.3) or end-organ dysfunction
- **Severe features:** BP ≥ 160/110, plt < 100k, LFT × 2, Cr > 1.1, pulmonary edema, headache / visual symptoms
- **sFlt-1/PlGF ratio** helps predict / rule out

Acute severe hypertension (≥ 160/110 for 15 min)

- **Labetalol IV**, **hydralazine IV** or **oral nifedipine** – treat within 30–60 min
- Target 140–150 / 90–100
- **Avoid:** ACEi / ARB, nitroprusside (cyanide)
- **Definitive treatment = delivery**

3. Eclampsia & Magnesium

- **$MgSO_4$:** 4–6 g IV over 15–20 min, then 1–2 g/h for 24 h postpartum (Magpie, Collaborative Eclampsia trials)
- **Therapeutic level:** 4–7 mEq/L (2–3.5 mmol/L)

Mg level (mEq/L)	Effect
7–10	Loss of patellar reflex
> 10	Respiratory depression
> 15–25	Cardiac arrest

Toxicity → stop infusion + calcium gluconate 10% 10 mL IV

- Monitor reflexes, RR, urine output (renal excretion)
- Recurrent seizure → extra 2 g bolus; avoid phenytoin as first-line

4. HELLP & Liver Disorders

- **HELLP:** Hemolysis (LDH > 600, schistocytes), ↑ Liver enzymes (AST > 70), Low Platelets (< 100k)
- Complications: DIC, hepatic hematoma / rupture, AKI, placental abruption
- **Rx:** stabilize, $MgSO_4$, BP control, **delivery**; platelets > 50k for cesarean

Acute fatty liver of pregnancy (AFLP)

- 3rd trimester; hypoglycemia, ↑ ammonia, ↑ bilirubin, coagulopathy, AKI
- **Swansea criteria**; LCHAD deficiency in fetus
- Rx: prompt delivery, supportive

Feature	HELLP	AFLP	TTP
Hypoglycemia	–	++	–
Hypertension	++	±	±
Coagulopathy (↓fibrinogen)	±	++	–
ADAMTS13 < 10%	–	–	++

5. Postpartum Hemorrhage

- **Definition:** ≥ 1000 mL blood loss or hypovolemia signs within 24 h
- **4 T's:** Tone (70%), Trauma, Tissue, Thrombin
- **Uterotonics:** oxytocin (1st) → ergometrine (avoid in HTN) → carboprost (avoid in asthma) → misoprostol
- **Tranexamic acid 1 g IV within 3 h** (WOMAN trial: ↓ death from bleeding)
- **Fibrinogen < 2 g/L** predicts severe PPH → cryoprecipitate / fibrinogen concentrate
- **Massive transfusion** 1:1:1; viscoelastic testing (ROTEM/TEG)
- Bakri balloon, B-Lynch suture, uterine artery embolization, **hysterectomy**

6. Antepartum Hemorrhage

- **Placenta previa:** painless bright bleeding; no digital exam
- **Placental abruption:** painful, tender hard uterus; DIC, fetal distress
- **Placenta accreta spectrum:** prior cesarean + previa → plan cesarean hysterectomy
- **Uterine rupture:** prior scar, sudden pain, fetal bradycardia

7. Amniotic Fluid Embolism

- Sudden **hypoxemia + hypotension / arrest + DIC** during labor or ≤ 30 min postpartum
- Biphasic: acute RV failure → LV failure
- **Clinical diagnosis** (no confirmatory test)
- Rx: supportive – airway, inotropes (RV support), aggressive blood products, early delivery
- A-OK (atropine, ondansetron, ketorolac): anecdotal

8. Cardiac Arrest in Pregnancy

- **Manual left uterine displacement** during CPR
- Standard ACLS drugs & doses; defibrillation is safe
- **Resuscitative hysterotomy** (perimortem cesarean) if no ROSC by **4 min → delivery by 5 min** (> 20 weeks)
- Early airway with smaller ETT (6–7)
- **Causes "BEAU-CHOPS":** Bleeding, Embolism, Anesthetic complications, Uterine atony, Cardiac, Hypertension, Other, Placental, Sepsis
- **Mg toxicity** → calcium; **LAST** (local anesthetic toxicity) → lipid emulsion 20%

9. Thromboembolism in Pregnancy

- Leading cause of maternal death in high-income countries
- **D-dimer** physiologically ↑ (YEARS algorithm adapted for pregnancy)
- **Imaging:** CUS first if leg symptoms; CXR → V/Q (normal CXR) or CTPA
- **Treatment: LMWH** (DOACs & warfarin teratogenic / contraindicated)
- Massive PE: thrombolysis acceptable if life-threatening

10. Peripartum Cardiomyopathy

- HF in last month of pregnancy to 5 months postpartum; EF < 45%
- **Rx:** standard HF therapy (no ACEi antepartum → hydralazine/nitrate), anticoagulation if EF < 35%
- **Bromocriptine** (blocks prolactin cleavage) – adjunct
- Severe: inotropes, MCS / ECMO, transplant
- Other cardiac: mitral stenosis decompensation, aortic dissection, MI

11. Maternal Sepsis

- Leading causes: **endometritis**, chorioamnionitis, pyelonephritis, wound infection, septic abortion
- Organisms: GAS (Streptococcus pyogenes – toxic shock), E. coli, anaerobes
- **Antibiotics within 1 h**; source control (evacuate retained products)
- Modified obstetric early warning scores (MEOWS)

12. Respiratory Issues

- **Asthma, pneumonia, influenza** – vaccinate; oseltamivir
- **Tocolytic pulmonary edema** (β-agonists, $MgSO_4$)
- **Ventilation:** target $PaCO_2$ 30–32, SpO_2 ≥ 95%, PaO_2 > 70 for fetus
- **Difficult airway** ↑ 8×; rapid desaturation
- ARDS: prone positioning & ECMO feasible

13. Drugs in Pregnancy

Safe	Avoid
Labetalol, hydralazine, nifedipine	ACEi, ARB, nitroprusside
LMWH, UFH	Warfarin, DOACs
Norepinephrine, phenylephrine	Valproate, phenytoin (if possible)
β-lactams, macrolides	Tetracyclines, fluoroquinolones, aminoglycosides
Insulin	Statins

- **Steroids** (betamethasone) if delivery < 34 weeks for fetal lung maturity
- **Anti-D** if Rh-negative after hemorrhage / trauma

14. Gynecologic Emergencies

- **Ruptured ectopic:** shock + positive β-hCG + free fluid → surgery
- **Ovarian hyperstimulation syndrome (OHSS):** ascites, hemoconcentration, VTE, AKI → fluids, albumin, paracentesis, LMWH
- **Ovarian torsion:** surgical emergency
- **Gestational trophoblastic disease:** hyperthyroidism, preeclampsia < 20 weeks
- **Trauma in pregnancy:** Kleihauer-Betke test, anti-D, fetal monitoring ≥ 4 h if > 20–24 weeks

15. High-Yield MCQ Pearls

- ☑ Normal $PaCO_2$ in pregnancy ≈ 30
- ☑ Severe HTN: labetalol / hydralazine / nifedipine within 60 min
- ☑ Eclampsia: $MgSO_4$ 4–6 g bolus + 1–2 g/h
- ☑ Lost reflexes → stop Mg; toxicity → calcium gluconate
- ☑ AFLP: hypoglycemia + coagulopathy
- ☑ TTP: ADAMTS13 < 10%
- ☑ PPH: TXA 1 g within 3 h
- ☑ Fibrinogen < 2 g/L → severe PPH
- ☑ Carboprost contraindicated in asthma
- ☑ Arrest: hysterotomy at 4–5 min
- ☑ LMWH for VTE in pregnancy
- ☑ Bromocriptine for peripartum cardiomyopathy

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